Fast National University of computing and Emerging Sciences
Peshawar Campus

Data Structures

Practice Problem(Queue)

1. Reverse a queue: Write a function to reverse the elements of a queue using a stack.

2. Check if a queue is a palindrome: Implement a function to check if the elements in a queue form a palindrome.

3. Sort a queue: Write a function to sort a queue in ascending order.

4. Interleave the first half with the second half: Given a queue of integers, interleave the first half with the second half of the queue.

5. Reverse the first K elements: Write a function that reverses the first K elements of a queue while maintaining the order of the remaining elements.

6. Merge two sorted queues: Given two sorted queues, merge them into a single sorted queue.

7. Check if two queues are identical: Implement a function that compares two queues and checks if they contain the same elements in the same order.

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8. Reverse alternate levels of a queue: Given a queue, reverse the elements at alternate levels.

9. Sort a queue using a stack: Write a function to sort a queue using an auxiliary stack.

10. Find the maximum element in a queue: Implement a function to find the maximum element in a queue without modifying the queue's order.

11. Find the minimum element in a queue: Write a function to find the minimum element in a queue.

12. Rotate a queue: Rotate the queue such that the front element moves to the back k times.

13. Check if a queue is sorted: Implement a function to check if the elements in a queue are sorted in non-decreasing order.

14. Merge two queues by alternating elements: Merge two queues by alternating elements from each queue until both are empty.

15. Find the sum of all elements in a queue: Write a function to calculate the sum of all elements in a queue.

16. Count occurrences of a specific element: Given a queue and an integer, count how many times the integer occurs in the queue.

17. Find the kth largest element in a queue: Write a function to find the kth largest element in a queue without changing the queue's order.

18. Find the kth smallest element in a queue: Write a function to find the kth smallest element in a queue.

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19. Rearrange a queue so that all even elements come before odd elements: Write a function to rearrange the queue such that all even elements come before odd elements, without changing the relative order of the even and odd elements.

20. Reverse a queue using recursion: Write a recursive function to reverse a queue.

21. Sort a queue using only recursion: Implement a recursive function to sort the elements of a queue.

22. Remove all occurrences of a specific element: Write a function to remove all occurrences of a given element from a queue.

23. Sum of alternating elements in a queue: Find the sum of alternating elements in a queue (i.e., sum elements at even positions).

24. Reverse the second half of a queue: Write a function that reverses the second half of a queue while keeping the first half unchanged.

25. Find the middle element of a queue: Write a function to find the middle element of a queue.

26. Split a queue into two equal halves: Implement a function to split a queue into two equal-sized queues.

27. Interleave even and odd elements of a queue: Rearrange the queue such that even and odd elements are interleaved.

28. Calculate the product of all elements in a queue: Write a function to calculate the product of all elements in a queue.

29. Move all zeros to the end: Write a function to move all zeros in a queue to the end while maintaining the order of non-zero elements.

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